

SCI225 Week 2 — Complete Study Guide

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Adaptive Immunity · Hypersensitivity (Types I–IV) · Autoimmunity · Primary & Secondary Immunodeficiency · Infection · Stress & Disease

Study Spine — for every immune topic ask:

1. Which arm of immunity is involved (antibody or T cell)?
2. What is being attacked?
3. How fast does the reaction develop?
4. What's the classic clinical clue?

MECHANISM

WHY

COMPENSATION

PATHOLOGY

MISSTEP

ACRONYM

EXAM TIP

Course alignment

This guide maps to the four Week 2 learning objectives:

- **LO1** — Describe hypersensitivity reactions, immune deficiencies, and their clinical consequences (Units 2, 4)
- **LO2** — Understand the process of adaptive immunity (Unit 1)
- **LO3** — Explain infection and defects in mechanisms of defense (Units 4, 5)
- **LO4** — Understand stress and the consequence of disease (Unit 6)

Every practice question here was written fresh for this guide. None reproduce exam items. The purpose is to *teach you to apply* the concept, then prove you can.

Unit 1 • Adaptive Immunity LO2

Adaptive immunity is **slow, specific, and has memory**. Unlike innate immunity (fast, nonspecific), adaptive takes days to mount on first exposure but responds within hours on re-exposure — that is the entire reason vaccination works.

MECHANISM Adaptive immunity has two arms that work together:

- **Humoral (B-cell) immunity** — B cells differentiate into plasma cells that make antibodies. Antibodies target *extracellular* threats: bacteria in the bloodstream, toxins, viruses before they enter cells.
- **Cell-mediated (T-cell) immunity** — T cells directly kill infected cells (CD8 cytotoxic) or coordinate the response (CD4 helper). Needed for *intracellular* pathogens: viruses already inside cells, intracellular bacteria (TB), tumor cells.

1.1 Cells of Adaptive Immunity

Cell	Origin	Main job	Key marker
B cell	Bone marrow (matures there)	Makes antibodies; becomes plasma cell or memory B cell	CD19, CD20
CD4 T helper	Bone marrow → thymus	Coordinates: secretes cytokines, activates B and CD8 cells	CD4

CD8 cytotoxic T cell	Bone marrow → thymus	Kills virus-infected and tumor cells directly (perforin, granzyme)	CD8
Regulatory T cell (T-reg)	Thymus	Suppresses self-reactive T cells → prevents autoimmunity	CD4, CD25, FoxP3
Plasma cell	Terminally differentiated B cell	Antibody factory — pumps out IgG (and others) for days	CD138
Memory B / T cell	Post-exposure B and T cells	Live for years; respond within hours on re-exposure	Varies

★ **EXAM TIP** T cells mature in the **T**hymus. B cells mature in the **B**one marrow. This single mnemonic saves a lot of confusion. A child born without a thymus (DiGeorge, Unit 4) has no functional T cells.

1.2 The Five Immunoglobulins — MADGE

ACRONYM / MNEMONIC MADGE — the five antibody classes:

IgM · IgA · IgD · IgG · IgE

Ig	Where / What	Role	Clinical clue
IgM	Pentamer; first to appear	Initial response, best complement activator	"Acute or recent infection" — rising IgM
IgG	Most abundant; crosses placenta	Long-term immunity, opsonization, memory	High IgG = past exposure or vaccination
IgA	Mucosal secretions: saliva, tears, breast milk, GI/respiratory	Protects mucosal surfaces; passes passive immunity through breast milk	Recurrent sinus/GI infections → IgA deficiency
IgE	Bound to mast cells	Type I hypersensitivity; parasitic infection	"Allergy, anaphylaxis, asthma, parasites"
IgD	On naïve B-cell surface	B-cell receptor; role poorly understood	Rarely tested clinically

WHY Why IgM comes first. B cells initially express IgM without needing help. With help from CD4 T cells, they can "class-switch" to IgG, IgA, or IgE depending on the signal — a more specialized, longer-lasting response. The first wave (IgM) buys time while the class-switched wave (IgG) builds.

1.3 Active vs Passive Immunity & Vaccines

MECHANISM ACTIVE IMMUNITY — you make it

Your own B/T cells mount a response. Slow onset (days), long-lasting (often lifelong), memory formed.

Natural: surviving infection.

Artificial: vaccination.

MECHANISM PASSIVE IMMUNITY — given to you

Pre-formed antibodies transferred in. Immediate (hours), short-lived (weeks to months), **no memory**.

Natural: maternal IgG across placenta, IgA in breast milk.

Artificial: IVIG, rabies immunoglobulin, antivenom, equine antiserum.

⚠ **COMMON MISSTEP** "Given" = **passive**. **Always**. If the patient received pre-formed antibody (RhoGAM, rabies Ig, monoclonal Ab, maternal IgG, breast milk IgA), they have passive immunity with no memory. If the patient *made* the antibody (infection, vaccine), it is active with memory.

Vaccine types — quick reference

Type	How it works	Examples	Key fact
Live attenuated	Weakened live pathogen	MMR, varicella, oral polio, BCG, yellow fever	Strong, long immunity; avoid in immunocompromised and pregnancy
Inactivated (killed)	Dead organism	Influenza (shot), hepatitis A, rabies, polio (IPV)	Safe for immunocompromised; needs boosters
Subunit / conjugate	Piece of antigen only	Hepatitis B, HPV, pertussis, pneumococcal	Safe, no live organism
Toxoid	Inactivated toxin	Tetanus, diphtheria	Creates antitoxin antibodies
mRNA	Lipid nanoparticle with mRNA → cell makes antigen	COVID-19 (Pfizer, Moderna)	No live virus; rapid design

★ **EXAM TIP** Live vaccines + immunocompromised = **danger**. An infant with profound lymphopenia and absent Igs (SCID suspect) who got a live vaccine is at risk of vaccine-derived disseminated infection. This is an exam favorite.

Practice Questions — Adaptive Immunity

Q1. Which statement best distinguishes humoral from cell-mediated immunity?

- A. Humoral targets intracellular pathogens; cell-mediated targets extracellular
- B. Humoral involves antibodies from B-cell-derived plasma cells; cell-mediated involves T cells directly killing infected cells
- C. Humoral is innate; cell-mediated is adaptive
- D. Humoral forms no memory; cell-mediated does

[Show answer](#)

B. Humoral = antibody = B cells (for extracellular threats). Cell-mediated = T cells (for intracellular and tumor cells). Both are part of adaptive immunity, both form memory.

Q2. A pediatric patient has recurrent sinus and gastrointestinal infections but normal IgG and IgM. Which immunoglobulin is most likely deficient?

- A. IgE
- B. IgA
- C. IgD
- D. IgM

[Show answer](#)

B. IgA lives in mucosal secretions and protects the respiratory and GI tracts. Selective IgA deficiency is the most common primary immunodeficiency and presents this way.

Q3. A newborn is born full-term to a fully vaccinated mother and is exclusively breastfed. The infant is protected from many pathogens during the first six months by:

- A. Natural active immunity from in utero infection
- B. Natural passive immunity from maternal IgG across the placenta and IgA in breast milk
- C. Artificial active immunity from vaccination at birth
- D. Innate immunity alone

[Show answer](#)

B. The infant "received" the antibodies → **passive**. The transfer was physiologic (placenta, breast milk) → **natural**. No memory is formed, which is why routine vaccination begins around 2 months.

Q4. Select all that apply. Which vaccines should be avoided in a patient with severe combined immunodeficiency (SCID)?

- A. MMR (measles, mumps, rubella)
- B. Hepatitis B
- C. Varicella
- D. Inactivated influenza
- E. Oral polio (OPV)

[Show answer](#)

A, C, E. All three are **live attenuated** — contraindicated in SCID because even a "weakened" live virus can cause disseminated infection when T cells are absent. Hep B (subunit) and inactivated influenza are safe.

Q5. A patient exposed to rabies receives both the rabies vaccine and rabies immunoglobulin. The immunoglobulin provides:

- A. Artificial active immunity
- B. Artificial passive immunity — immediate but short-lived
- C. Natural passive immunity
- D. Long-term cellular memory

[Show answer](#)

B. Pre-formed antibody given = artificial passive = immediate protection, no memory. The vaccine provides the artificial *active* component for memory. Post-exposure rabies is a classic case for using both together.

Unit 2 • Hypersensitivity Types I–IV LO1

Hypersensitivity = an immune response that harms the host. The four Gell & Coombs types are grouped by *mechanism*, not by symptom. Learn the mechanism cold — the exam vignettes vary but the mechanism is fixed.

ACRONYM / MNEMONIC "ACID" — the four hypersensitivity types, alphabetized by what drives them:

Allergic · Cytotoxic · Immune complex · Delayed

Or: I = IgE · II = IgG/IgM on a cell surface · III = immune complexes · IV = T cells (no antibody).

Type	Driver	Timing	Classic examples
I — Immediate / anaphylactic	IgE on mast cells	Minutes	Anaphylaxis (food, bee sting, latex), allergic rhinitis, atopic asthma, eczema
II — Cytotoxic / antibody-mediated	IgG or IgM binds cell-surface antigen	Minutes to hours	ABO transfusion reaction, hemolytic disease of newborn, Graves, myasthenia gravis, rheumatic fever, hyperacute transplant rejection
III — Immune complex	Antigen-antibody complexes deposit in tissues → complement activation	1–3 weeks	SLE/lupus nephritis, serum sickness, farmer's lung, post-strep glomerulonephritis, Arthus reaction
IV — Delayed / cell-mediated	Sensitized T cells (no antibody)	24–72 hours	TB skin test (PPD), contact dermatitis (poison ivy, nickel, latex), chronic transplant rejection, graft-versus-host, type 1 DM

2.1 Type I — Immediate / IgE-Mediated

MECHANISM First exposure (sensitization): allergen → B cells make IgE → IgE binds Fc receptors on mast cells and basophils. No symptoms yet.

Second exposure: allergen cross-links two adjacent IgE → mast cell **degranulates** → histamine + leukotrienes + tryptase released → bronchoconstriction, vasodilation, ↑ capillary permeability.

Allergen (2nd exposure) → cross-links IgE on mast cell → degranulation → histamine release → smooth muscle contraction + vasodilation + capillary leak → wheezing, hives, hypotension

PATHOLOGY Anaphylaxis is systemic type I — throat/bronchus constriction (airway), hypotension from vasodilation (shock), urticaria/angioedema. Treat with **IM epinephrine** first (α = vasoconstriction, β_1 = cardiac output, β_2 = bronchodilation), then antihistamines + steroids as adjuncts.

★ **EXAM TIP** Timing is the fastest clue. If symptoms appear within **minutes** and the patient has a prior exposure, think Type I. Bee stings, peanuts, shellfish, pollen, latex, cats, and medications are the classic triggers.

2.2 Type II — Cytotoxic / Antibody-Mediated

MECHANISM IgG or IgM binds an antigen that is *attached to a cell surface*. This triggers: complement activation (cell lysis), phagocytosis, NK-cell killing (antibody-dependent cellular cytotoxicity), or altered cell function (blocking or stimulating receptors).

The four flavors of type II — same mechanism, different tissue targets:

- **Direct cell destruction** — ABO transfusion reaction (anti-A or anti-B antibodies lyse the wrong-type RBCs within minutes)
- **Hemolytic disease of the newborn** — Rh-negative mom who was previously sensitized makes IgG anti-Rh → crosses placenta → attacks Rh-positive fetal RBCs → fetal anemia and jaundice. **RhoGAM** prevents sensitization.
- **Receptor stimulation** — **Graves disease**: IgG binds TSH receptor on thyroid and *activates* it → hyperthyroidism
- **Receptor blockade** — **myasthenia gravis**: IgG blocks ACh receptors at the neuromuscular junction → fatigable muscle weakness
- **Molecular mimicry** — **rheumatic fever**: antibodies made against strep M protein cross-react with heart, joints, and basal ganglia → carditis, arthritis, chorea

PATHOLOGY Hyperacute transplant rejection is type II: recipient had pre-formed antibodies (from prior transplant, transfusion, or pregnancy) against donor HLA → graft is attacked within *minutes of reperfusion* → becomes pale and nonfunctioning. A positive pre-transplant crossmatch should prevent this.

2.3 Type III — Immune Complex

MECHANISM Antigen and antibody form **soluble immune complexes** in the blood. Complexes deposit in small vessels and filtration organs — especially kidneys (glomeruli), joints, and skin. Complement is activated locally → neutrophil recruitment → tissue damage.

High-yield type III diseases:

- **SLE (systemic lupus erythematosus)** — anti-dsDNA and anti-Sm antibodies form complexes that deposit in glomeruli (lupus nephritis), joints, and skin. Malar butterfly rash, proteinuria, hematuria, low complement.
- **Serum sickness** — exposure to a foreign protein (classically equine antiserum) → immune complexes form 7–14 days later → fever, rash, joint pain, proteinuria. Self-resolving.
- **Post-streptococcal glomerulonephritis** — strep antigens form complexes that deposit in glomeruli 1–3 weeks after infection → hematuria, edema, HTN.
- **Farmer's lung / hypersensitivity pneumonitis** — inhaled moldy hay antigens → complexes in lung → dyspnea, cough hours after exposure.
- **Arthus reaction** — localized type III at a vaccine injection site: erythema and swelling hours after boosters from preformed antibody + local antigen.

★ **EXAM TIP** "Granular" deposits on immunofluorescence = immune complexes = type III. If the stem says "linear deposits," think Goodpasture (which is type II, anti-GBM).

2.4 Type IV — Delayed / T-Cell-Mediated

MECHANISM No antibody involved. Sensitized CD4 T cells release cytokines 24–72 hours after re-exposure to an antigen, recruiting macrophages that cause tissue damage. This is the slowest hypersensitivity by design — not because it's mild, but because it requires T cells to migrate and activate.

Classic type IV examples:

- **PPD (tuberculin skin test)** — positive reaction is firm induration at 48–72 h in a TB-sensitized patient
- **Contact dermatitis** — poison ivy (urushiol), nickel jewelry, latex, cosmetics, topical medications. Itchy vesicles 48–72 h after re-exposure.
- **Chronic transplant rejection** — T cells target donor HLA over months to years; gradual graft dysfunction.
- **Graft-versus-host disease** — donor T cells attack the recipient (opposite of rejection).
- **Type 1 diabetes** — T cells destroy pancreatic β -cells.
- **Multiple sclerosis, contact-type drug reactions.**

⚠ **COMMON MISSTEP** Poison ivy is **NOT** Type I. Students guess "allergic = IgE = Type I" reflexively. Poison ivy is delayed (hours to days), itch-predominant, T-cell mediated → **Type IV**. Same for nickel contact dermatitis and PPD tests.

2.5 Transplant Rejection — The Three Timelines

Rejection	Timing	Mechanism	Clinical
Hyperacute	Minutes of reperfusion	Type II — preformed recipient antibody vs donor HLA or ABO	Graft pale, mottled, nonfunctioning; immediate loss
Acute	Days to months	Type IV — recipient T cells attack donor cells	Graft tenderness, $\uparrow$ creatinine, responds to immunosuppression
Chronic	Months to years	Type IV (mostly) + vascular antibody damage — fibrosis and vessel narrowing	Gradual graft dysfunction; poor response to treatment

Practice Questions — Hypersensitivity

Q6. A 7-year-old with a known cat allergy develops wheezing and hives within 5 minutes of entering a home with cats. Which mechanism is most responsible?

- A. IgG and complement deposit in tissues
- B. Cross-linking of IgE on mast cells triggers histamine release
- C. CD8 T cells kill cat-antigen-presenting cells
- D. Immune complexes form in serum

[Show answer](#)

B. Rapid onset wheezing + hives in a sensitized patient = type I. Mast-cell degranulation releases histamine and leukotrienes that contract bronchial smooth muscle and raise capillary permeability.

Q7. A patient receives the wrong ABO blood type and immediately develops fever, flank pain, and hemoglobinuria. Which hypersensitivity type?

- A. Type I — IgE
- B. Type II — cytotoxic, IgG/IgM vs RBC surface antigens
- C. Type III — immune complex
- D. Type IV — delayed

[Show answer](#)

B. Anti-A or anti-B antibodies in the recipient bind to donor RBCs → complement lyses them → intravascular hemolysis (fever, flank pain, hemoglobinuria = free Hb filtered into urine). Type II is always antibody directed at a cell surface antigen.

Q8. Matching. Match each clinical scenario to the hypersensitivity type.

Scenario	Type
Anaphylaxis to peanuts	___
Hemolytic disease of the newborn	___
Lupus nephritis	___
Poison ivy contact dermatitis	___
Serum sickness after equine antiserum	___
Positive PPD test	___

Choices: I · II · III · IV

[Show answer](#)

Peanut anaphylaxis → I · HDN → II · Lupus nephritis → III · Poison ivy → IV · Serum sickness → III · PPD → IV. Remember: timing is your first clue — minutes (I, II), 1–3 weeks (III), 24–72 h (IV).

Q9. A nursing student develops a firm, raised, red induration 48 hours after a TB skin test. Which hypersensitivity type?

- A. Type I — IgE-mediated
- B. Type II — cytotoxic
- C. Type III — immune complex
- D. Type IV — T-cell mediated

[Show answer](#)

D. 48 h delay + induration + no swelling/pruritus = T-cell response, not antibody. This is textbook type IV. Positive PPD indicates prior TB exposure and sensitized T cells.

Q10. Order. Place in correct sequence the events of a Type I anaphylactic reaction to peanuts in a previously sensitized child:

- Histamine, tryptase, and leukotrienes are released
- Peanut allergen cross-links IgE on mast cell surface
- Bronchoconstriction, vasodilation, and capillary leak occur
- Mast cell degranulation

[Show answer](#)

Correct order: Peanut allergen cross-links IgE on mast cells → mast cell degranulation → histamine/tryptase/leukotrienes released → bronchoconstriction, vasodilation, capillary leak. The whole cascade takes minutes.

Q11. A kidney transplant patient develops a pale, nonfunctioning graft within 10 minutes of reperfusion. Pre-transplant crossmatch had been positive. Which mechanism?

- A. Acute rejection — T cells attack donor HLA
- B. Chronic rejection — fibrosis of vessels
- C. Hyperacute rejection — preformed recipient antibody against donor antigen (Type II)
- D. Graft-versus-host disease

[Show answer](#)

C. Minutes + preformed antibody + positive crossmatch = hyperacute rejection. Prevention requires careful ABO/HLA matching and crossmatching before transplant.

Q12. Select all that apply. Which are type III hypersensitivity reactions?

- A. Systemic lupus erythematosus (SLE) nephritis
- B. Serum sickness after equine antitoxin
- C. Poison ivy dermatitis
- D. Hypersensitivity pneumonitis (farmer's lung)
- E. Anaphylaxis to bee sting

[Show answer](#)

A, B, D. All involve immune-complex deposition. Poison ivy is type IV; anaphylaxis is type I.

Q13. A 33-year-old with Graves disease has weight loss, tachycardia, and high T4. The pathophysiology is:

- A. Type I — IgE-mediated destruction of thyroid
- B. Type II — IgG binds and stimulates the TSH receptor
- C. Type III — immune complexes in thyroid
- D. Type IV — T cells destroy β -cells

[Show answer](#)

B. Graves is a type II hypersensitivity of receptor *stimulation*. The IgG mimics TSH and drives the thyroid to overproduce hormone. Contrast with Hashimoto (type IV, T-cell destruction of thyroid → hypothyroid).

Unit 3 • Autoimmunity LO1

Autoimmunity = immune system attacks the body's own tissues because it failed to distinguish self from non-self. Mechanisms overlap heavily with hypersensitivity types II, III, and IV — in fact, most named autoimmune diseases ARE one of those types targeting a specific tissue.

MECHANISM How tolerance breaks down:

- **Molecular mimicry** — a pathogen antigen resembles a self-antigen; antibodies cross-react (rheumatic fever after strep).
- **Loss of regulatory T cells** — fewer T-regs means self-reactive lymphocytes escape suppression.
- **Release of sequestered antigens** — trauma exposes antigens normally hidden from the immune system (sympathetic ophthalmia after eye injury).
- **Genetic predisposition** — certain HLA types raise risk (HLA-B27 in ankylosing spondylitis, HLA-DR3/DR4 in T1DM).

High-yield autoimmune diseases at a glance

Disease	Target	Mechanism	Clinical hallmarks
Systemic lupus erythematosus (SLE)	DNA, cell nuclei, many tissues	Type III (immune complex) — anti-dsDNA, anti-Sm	Malar butterfly rash, joint pain, lupus nephritis (proteinuria, hematuria), fatigue, low complement
Rheumatoid arthritis	Synovial joints	Type III + Type IV	Symmetric polyarthritis (MCP, PIP, wrist), morning stiffness > 1 hr, rheumatoid factor, anti-CCP
Myasthenia gravis	ACh receptor at NMJ	Type II — IgG blocks the receptor	Fatigable weakness worsening with use, ptosis, diplopia, positive AChR antibody test
Graves disease	TSH receptor on thyroid	Type II — IgG stimulates the receptor	Hyperthyroidism, goiter, exophthalmos, pretibial myxedema
Hashimoto thyroiditis	Thyroid follicular cells	Type IV — lymphocytic destruction	Hypothyroidism, goiter, anti-TPO antibodies
Type 1 diabetes	Pancreatic β -cells	Type IV — T cells destroy β -cells	Young onset, absolute insulin deficiency, DKA-prone
Rheumatic fever	Heart valves, joints, skin, brain	Type II — antibodies vs strep M protein cross-react (molecular mimicry)	2 weeks post strep: carditis (mitral regurg), arthritis, chorea, erythema marginatum, subcutaneous nodules
Multiple sclerosis	CNS myelin	Type IV — T cells demyelinate	Relapsing neuro deficits across "time and space" — optic neuritis, weakness, sensory loss
Goodpasture	Basement membrane of lung alveoli + renal glomeruli	Type II — anti-GBM antibodies (linear deposits)	Hemoptysis + hematuria

WHY Why women get most autoimmune disease. SLE, RA, MS, Graves, Hashimoto, and myasthenia all skew female — estrogen is immunostimulatory, X chromosome carries many immune genes, and women mount stronger immune responses overall. This helps pregnancy (mom's immunity tolerates the fetus) but raises autoimmune risk.

⚠ **COMMON MISSTEP Graves vs Hashimoto.** Same gland, opposite directions.
 Graves = IgG **stimulates** TSH receptor → **hyper**thyroid. Type II.
 Hashimoto = T cells **destroy** the gland → **hypo**thyroid. Type IV.

★ **EXAM TIP "Molecular mimicry" keywords** on the exam: "rheumatic fever after strep," "antibodies cross-react with heart." These point to type II autoimmunity triggered by infection.

Practice Questions — Autoimmunity

Q14. A patient presents with fatigable muscle weakness worsening with use, ptosis, and diplopia. Which antibody is most likely positive?

- A. Anti-dsDNA
- B. Anti-ACh receptor
- C. Anti-TPO
- D. Anti-CCP

[Show answer](#)

B. Myasthenia gravis — IgG blocks ACh receptors at the NMJ, so muscles fatigue quickly with repeated use (type II blockade).

Q15. A patient with SLE develops proteinuria, hematuria, and granular immune complex deposits on kidney biopsy. The hypersensitivity type is:

- A. Type I
- B. Type II
- C. Type III
- D. Type IV

[Show answer](#)

C. Lupus nephritis is the textbook type III autoimmune disease. "Granular" = immune complex; "linear" = type II anti-GBM (Goodpasture).

Q16. A child develops rheumatic fever two weeks after an untreated strep throat. Which principle best explains this?

- A. Direct bacterial invasion of the heart
- B. Molecular mimicry — antibodies against strep M protein cross-react with cardiac tissue
- C. IgE-mediated response to strep antigen
- D. Immune complex deposition in joints only

[Show answer](#)

B. The antibodies made against strep are the same antibodies now attacking heart, joints, skin, and basal ganglia. This is the classic molecular-mimicry type II autoimmune reaction.

Q17. Select all that apply. Which findings are typical of SLE?

- A. Malar "butterfly" rash
- B. Proteinuria from lupus nephritis
- C. Positive anti-dsDNA antibody
- D. High serum complement
- E. Arthralgia and fatigue

[Show answer](#)

A, B, C, E. D is wrong — SLE *consumes* complement (C3 and C4 are typically **low** during active disease as immune complexes activate the cascade).

Unit 4 • Immunodeficiencies LO1, LO3

MECHANISM PRIMARY (congenital)

You are born with it. Usually genetic.
Presents in infancy or childhood with recurrent infections.

MECHANISM SECONDARY (acquired)

Caused by something else. Most common cause worldwide = **HIV/AIDS**. Others: malnutrition, chemotherapy, steroids, aging, splenectomy, diabetes, CKD.

4.1 Primary Immunodeficiencies

Disease	Defect	Arm affected	Clinical clue
Selective IgA deficiency	Low/absent IgA	Humoral (mucosal)	Recurrent sinus and GI infections; most common primary immunodeficiency; anaphylaxis to blood products containing IgA
X-linked (Bruton's) agammaglobulinemia	BTK mutation; no mature B cells	Humoral (all Igs)	Boys; recurrent bacterial infections after ~6 months (maternal IgG wanes); low/absent all immunoglobulins
Common variable immunodeficiency (CVID)	Poor antibody production	Humoral	Adult onset (20s–30s), recurrent sinopulmonary infections, ↑ autoimmunity and lymphoma risk
DiGeorge syndrome (22q11.2 deletion)	Thymic and parathyroid hypoplasia	Cellular (T cells)	"CATCH-22": Cardiac defects, Abnormal facies, Thymic aplasia, Cleft palate, Hypocalcemia (tetany); 22q deletion
SCID (severe combined immunodeficiency)	T and B cell defects (multiple genes, e.g. adenosine deaminase)	Both	Infant with failure to thrive, chronic diarrhea, opportunistic infections, profound lymphopenia; live vaccines deadly
Complement deficiency (C5–C9)	Late complement components	Complement	Recurrent <i>Neisseria</i> infections (meningococcal, gonococcal)

ACRONYM / MNEMONIC CATCH-22 for DiGeorge syndrome (chromosome 22q11.2):

Cardiac defects · **A**bnormal facies · **T**hymic aplasia · **C**left palate · **H**ypocalcemia (from parathyroid aplasia → tetany).

★ **EXAM TIP** The pattern of infection tells you which arm is broken:

- Recurrent pyogenic bacteria (Strep, Staph, H. flu) → **B-cell / antibody** problem
- Opportunistic viruses, fungi, *Pneumocystis* → **T-cell** problem
- Recurrent *Neisseria* → **complement** problem
- Bacterial infections at skin and mucous membranes → **phagocyte** problem

4.2 Secondary Immunodeficiency — HIV / AIDS

MECHANISM HIV is a retrovirus that binds **gp120** → **CD4 receptor** on helper T cells → enters and integrates into the host genome. Over years, CD4 count drops. When $CD4 < 200/mm^3$ or an AIDS-defining opportunistic infection/malignancy occurs, the patient has **AIDS**.

AIDS-defining conditions (partial list): *Pneumocystis jirovecii* pneumonia, esophageal candidiasis, Kaposi sarcoma, CMV retinitis, cryptococcal meningitis, toxoplasmosis, TB.

CD4 range	Risk
> 500/mm ³	Normal immune function
200–500	Early symptoms: thrush, shingles, TB reactivation
50–200	AIDS-defining: PCP, candidiasis (esophageal), Kaposi
< 50	End-stage: CMV, MAC, CNS toxoplasmosis, lymphoma

PATHOLOGY Why HIV is so destructive. CD4 helpers coordinate BOTH arms of adaptive immunity. Lose them and the humoral arm weakens (B cells don't class-switch well), and the cellular arm collapses (CD8 T cells need CD4 help). Innate immunity and mucosal barriers remain — but once those fail, opportunistic infections take over.

⚠ **COMMON MISSTEP** "HIV kills the patient." No — HIV destroys CD4 cells. *Opportunistic infections* (PCP, cryptococcus, CMV) are what kill AIDS patients.

Practice Questions — Immunodeficiency

Q18. An infant with DiGeorge syndrome has tetany, hypocalcemia, and recurrent viral infections. The tetany is caused by:

- A. Thymic aplasia → T-cell deficiency
- B. Parathyroid aplasia → hypocalcemia
- C. Cardiac defect → low cardiac output
- D. Cleft palate → feeding difficulty

Show answer

B. DiGeorge = 22q11.2 deletion → thymic AND parathyroid hypoplasia. Low Ca^{2+} raises nerve excitability → tetany. Viral infections are from the separate T-cell problem (thymic aplasia).

Q19. An adolescent has recurrent sinusitis and GI infections with undetectable IgA but normal IgG and IgM. The most likely diagnosis is:

- A. SCID
- B. X-linked agammaglobulinemia
- C. Selective IgA deficiency
- D. HIV infection

Show answer

C. Isolated low IgA + mucosal infections = selective IgA deficiency. SCID has both T and B deficits; Bruton's lacks all immunoglobulins; HIV shows CD4 loss. Selective IgA is the most common primary immunodeficiency.

Q20. A patient with recurrent *Neisseria meningitidis* infections and normal immunoglobulins has which most likely defect?

- A. B-cell deficiency
- B. T-cell deficiency
- C. Phagocyte defect
- D. Complement deficiency (late components C5–C9)

[Show answer](#)

D. Late complement (C5–C9) forms the membrane attack complex needed for lysing *Neisseria*. Without it, these Gram-negative cocci cause recurrent disease even with normal antibodies.

Q21. A patient with HIV has oral thrush, a CD4 of 180/mm³, and elevated viral load. Which condition does the patient have?

- A. HIV infection without AIDS
- B. AIDS (CD4 < 200 or AIDS-defining illness)
- C. Primary immunodeficiency
- D. Acute HIV seroconversion

[Show answer](#)

B. CD4 < 200 defines AIDS regardless of symptoms. Oral candidiasis confirms the cellular immunity is already impaired. ART is started immediately.

Q22. An infant with profound lymphopenia and absent IgM and IgA was just given a live vaccine. The most dangerous risk is:

- A. Type I allergic reaction to the vaccine
- B. Disseminated vaccine-derived infection from the live organism
- C. Local pain at injection site
- D. Loss of future memory immunity

[Show answer](#)

B. SCID is suspected. Without T and B cells, the attenuated organism can replicate unchecked and cause systemic disease. Live vaccines are absolutely contraindicated in SCID.

Q23. Select all that apply. Which conditions cause secondary (acquired) immunodeficiency?

- A. Long-term corticosteroid therapy
- B. HIV infection
- C. Chemotherapy
- D. Severe malnutrition
- E. DiGeorge syndrome
- F. Splenectomy

[Show answer](#)

A, B, C, D, F. DiGeorge (E) is primary/congenital. Everything else acquires the deficiency secondarily.

Unit 5 • Infection Process LO3

5.1 The Four Stages of Clinical Infection

Incubation → Prodromal → Illness (acute) → Convalescence

Stage	What's happening	Key features
1. Incubation	Pathogen multiplies but no symptoms yet	Patient may be contagious; length varies (common cold 1–3 days, HIV years)
2. Prodromal	Vague, early symptoms as immune response begins	Fatigue, low-grade fever, malaise; patient is highly contagious
3. Illness (acute)	Full clinical disease	Characteristic symptoms; peak pathology; immune response in full swing
4. Convalescence	Recovery as pathogen is cleared	Symptoms resolve; memory forming; possible post-infectious complications (GN, rheumatic fever)

5.2 How Pathogens Cause Injury

- **Direct tissue invasion** — bacteria or viruses physically enter and damage cells (Shigella, influenza).
- **Exotoxins** — proteins *secreted* by (usually Gram-positive) bacteria. Specific, potent, often heat-labile. Tetanus toxin, botulinum toxin, diphtheria toxin, cholera toxin.
- **Endotoxins** — lipopolysaccharide (LPS) in the Gram-negative outer membrane, *released when bacteria lyse*. Drives septic shock via TNF- α , IL-1.
- **Immune evasion** — capsules (pneumococcus), antigenic variation (HIV, influenza), intracellular hiding (TB, listeria), IgA protease (gonorrhea).
- **Adherence** — pili, fimbriae let bacteria stick to mucosa.
- **Trigger pathologic immune response** — superantigens (toxic shock), molecular mimicry (rheumatic fever).

★ **EXAM TIP** Exo vs endo toxin — the simple split:

Exotoxin = **ex**creted by bacteria, mostly Gram-**positive**, protein, heat-**labile**, often **vaccine-preventable** (tetanus, diphtheria toxoid vaccines).

Endotoxin = **i**nside the cell wall, Gram-**negative**, LPS, heat-**stable**, no specific vaccine, drives septic shock.

⚠ **COMMON MISSTEP** "The patient is febrile but not contagious." Usually wrong during infection stages. Prodromal is when patients are often *most* contagious because they don't know they're sick. Practice public-health reasoning: "when do we isolate?"

Practice Questions — Infection

Q24. Order. Place the stages of clinical infection in the correct sequence:

- Illness stage with full characteristic symptoms
- Convalescence
- Incubation period
- Prodromal stage

[Show answer](#)

Correct order: Incubation → Prodromal → Illness → Convalescence. Memorize this and the fact that patients are often contagious during incubation and especially during prodromal stages.

Q25. Select all that apply. Which are mechanisms pathogens use to cause infection or cell injury?

- A. Release of exotoxins
- B. Endotoxin-triggered cytokine storm
- C. Direct cell invasion and lysis
- D. Antigenic variation to evade antibodies
- E. Production of IgA protease
- F. Activation of host regulatory T cells

[Show answer](#)

A, B, C, D, E. F is protective, not harmful — pathogens don't usually boost T-reg function to cause disease. The rest are classic virulence mechanisms.

Q26. A student is learning about vaccines. Which statement indicates a need for further teaching?

- A. "Live attenuated vaccines give the strongest immunity but are avoided in immunocompromised patients."
- B. "Toxoid vaccines contain inactivated toxins that create antitoxin antibodies."
- C. "Passive immunity from IVIG gives immediate but temporary protection with no memory."
- D. "An mRNA vaccine works by inserting a piece of viral DNA permanently into my genome."

[Show answer](#)

D. mRNA vaccines deliver mRNA (not DNA) that is translated transiently in the cytoplasm and degraded — it does NOT integrate into DNA. The other statements are correct.

Q27. A patient in septic shock from a Gram-negative bacteremia has hypotension and a rising lactate. Which toxin is most responsible for the shock?

- A. Tetanus exotoxin
- B. Lipopolysaccharide (LPS) endotoxin
- C. Diphtheria exotoxin
- D. Botulinum exotoxin

[Show answer](#)

B. Gram-negative septic shock = endotoxin (LPS) → massive TNF- α /IL-1 release → vasodilation, $\uparrow$ capillary permeability, hypotension. The other three are exotoxins from Gram-positive organisms.

Unit 6 • Stress & Immunity LO4

MECHANISM Chronic psychological stress → activation of the HPA axis → sustained elevation of **cortisol**. Cortisol has many actions, but the relevant ones here are:

- Suppression of lymphocyte proliferation and cytokine release
- Redistribution of immune cells away from circulating blood

- Decreased antibody production
- Impaired wound healing (↓ collagen synthesis, ↓ fibroblast activity)

The result: more infections, slower wound healing, reactivation of latent viruses (HSV, VZV), and poorer response to vaccines.

Chronic stress → HPA axis → ↑ cortisol → ↓ lymphocyte function + ↓ antibodies + ↓ wound healing → ↑ infection, slow healing

Conditions and states that weaken host defense (acquired / situational)

- **Chronic psychological stress** — cortisol-mediated immunosuppression
- **Severe malnutrition** — protein-energy deficit → impaired T and B cell function
- **Sleep deprivation** — reduces NK cell activity, T-cell responses
- **Diabetes mellitus** — hyperglycemia impairs neutrophil chemotaxis and phagocytosis
- **Advanced age** — immunosenescence
- **Chronic alcohol use** — impairs multiple immune pathways; worsens malnutrition
- **Chemotherapy / immunosuppressants / corticosteroids** — direct suppression
- **Chronic illness (CKD, cirrhosis, heart failure)** — multiple pathways
- **Splenectomy** — loss of defense against encapsulated organisms (pneumococcus, H. flu, meningococcus)

WHY Why wound healing slows under stress. Cortisol suppresses fibroblasts, reduces collagen synthesis, and dampens the inflammatory response needed for tissue repair. A chronically stressed surgical patient heals more slowly and is more vulnerable to wound infection than a calm one.

★ **EXAM TIP Stress + infection + slow healing** — when a vignette lists all three, the answer is almost always "chronic stress → cortisol → immunosuppression."

Practice Questions — Stress & Immunity

Q28. A patient under prolonged psychological stress develops frequent infections and a wound that heals slowly. Which explanation best accounts for this?

- Stress raises histamine, triggering allergies
- Chronic stress elevates cortisol, which suppresses lymphocyte function and impairs wound healing
- Stress increases complement activation
- Stress only affects psychological health, not immunity

[Show answer](#)

B. Chronic HPA axis activation → ↑ cortisol → immunosuppression. Acute stress is mildly immune-boosting; it's the *chronic* pattern that impairs defense.

Q29. Select all that apply. Which states weaken host defenses and raise infection risk?

- Chronic sleep loss
- Poorly controlled diabetes
- Daily aerobic exercise of moderate intensity
- Protein-energy malnutrition

- E. Prolonged high-dose corticosteroid therapy
- F. Splenectomy

[Show answer](#)

A, B, D, E, F. Moderate regular exercise actually *improves* immune function; all the others impair it. Splenectomized patients need pneumococcal, meningococcal, and H. flu vaccines.

Q30. A 72-year-old on chronic prednisone for rheumatoid arthritis develops shingles. Which mechanism best explains reactivation?

- A. Direct antiviral effect of prednisone
- B. Corticosteroid suppression of T-cell surveillance allowing latent VZV to reactivate
- C. New primary varicella infection
- D. Allergic reaction to the medication

[Show answer](#)

B. T-cell surveillance normally holds latent viruses (HSV, VZV, CMV, EBV) in check. Steroid suppression lets VZV reactivate along a dermatome → shingles. Aging further increases risk — shingles vaccine is recommended age 50+.

Appendix • Top Missteps & Master Mnemonics

Top 15 Week 2 Missteps

#	The trap	What's actually true
1	"Poison ivy is type I allergy."	It's type IV — delayed T-cell. 48–72 h onset, no IgE involved.
2	"Graves and Hashimoto are both type II."	Graves is type II (IgG stimulates TSH-R). Hashimoto is type IV (T-cell destruction). Opposite gland outcomes.
3	"Given antibodies = active immunity."	Given = passive , always. No memory.
4	"SLE has high complement."	SLE has low C3/C4 — immune complexes consume complement during flares.
5	"IgE is most abundant."	IgG is. IgE is the least abundant and is the allergy / parasite antibody.
6	"Hyperacute rejection is T-cell mediated."	It's antibody-mediated type II (preformed antibodies vs donor HLA or ABO). T-cell = acute/chronic rejection.
7	"HIV directly kills the patient."	HIV depletes CD4. Opportunistic infections kill AIDS patients.
8	"Live vaccines are safe for everyone."	Contraindicated in SCID, AIDS, pregnancy, and chemotherapy.
9	"CD4 < 500 = AIDS."	AIDS is CD4 < 200 <i>or</i> an AIDS-defining illness.
10	"Rheumatic fever is direct strep invasion of the heart."	It's molecular mimicry — antibodies to strep M protein cross-react with heart.
11		

	"ABO transfusion reaction is anaphylaxis."	It's type II cytotoxic (antibodies vs RBC surface antigens). Anaphylaxis requires IgE.
12	"Selective IgA deficiency is rare."	It is the most common primary immunodeficiency.
13	"Acute stress weakens immunity."	Acute stress is mildly immune-boosting. Only chronic stress → cortisol → immunosuppression.
14	"Exotoxins are Gram-negative."	Exotoxins = mostly Gram- positive . Endotoxin (LPS) = Gram-negative.
15	"Myasthenia gravis is receptor stimulation."	MG is receptor blockade . Graves is the stimulation one.

Master Mnemonic Sheet

Mnemonic	Meaning
MADGE	The 5 immunoglobulin classes: IgM, IgA, IgD, IgG, IgE
ACID	4 hypersensitivity types: Allergic, Cytotoxic, Immune complex, Delayed
I (IgE) · II (IgG on cell) · III (complex) · IV (T cell)	Mechanism by type
CATCH-22	DiGeorge: Cardiac, Abnormal facies, Thymic aplasia, Cleft palate, Hypocalcemia, chr 22q11.2
T in Thymus, B in Bone marrow	Where each cell matures
"Linear = Goodpasture"	Linear IF = type II anti-GBM; granular = type III immune complex
"Given" = Passive	Pre-formed antibody transfer — no memory
CD4 < 200 = AIDS	Cutoff regardless of symptoms
"Exo = excreted, Endo = inside wall"	Bacterial toxin split
"Minutes, hours, days, weeks"	Approximate timing of hypersensitivity types I, II, IV, III respectively

Quick Reference — Which arm is broken?

Recurrent infection pattern	Likely defect
Pyogenic bacteria (Strep, Staph, H. flu)	B-cell / antibody
Viruses, fungi, <i>Pneumocystis</i> , opportunistic	T-cell
Recurrent <i>Neisseria</i>	Late complement (C5–C9)
Skin and mucosal bacterial infections	Phagocyte
Sinopulmonary + GI, isolated	IgA deficiency
Encapsulated organisms	Splenectomy or sickle cell (functional asplenia)

Week 2 Study Strategy

- **Own the hypersensitivity types cold.** If you can name the mechanism and one classic example for all four, you have the majority of Week 2's exam points.
- **Learn by timing.** Minutes → Type I or hyperacute II. Hours → exotoxin, sepsis. 24–72 h → Type IV. Weeks → Type III, post-infectious autoimmunity.
- **Pair autoimmune disease to its hypersensitivity type.** Every autoimmune disease in this week is one of the four types — know which.
- **Match infection pattern → defective arm.** If a vignette describes recurrent *Neisseria*, think complement. Recurrent viruses and fungi, think T-cells.
- **Don't mix Graves and Hashimoto.** Same organ, opposite direction, opposite hypersensitivity type.
- **Respect the CD4 numbers.** 500 = normal floor. 200 = AIDS threshold. 50 = end-stage. These get asked directly.
- **Live vaccine + immunocompromise = no.** If you see SCID, HIV advanced, chemotherapy, pregnancy, or organ transplant + a live vaccine option — it's the wrong answer.

SCI225 Week 2 Study Guide — built from the Week 2 learning objectives covering adaptive immunity, hypersensitivity (Types I–IV), autoimmunity, primary and secondary immunodeficiency, infection, and stress. Made by Dr. Vidya Aggrawal to teach the concepts tested, not the exam questions themselves. Uses the same color-coded format as the Week 1 guide and the midterm review. Print for a paper copy, or use in-browser with collapsible practice answers.